Jingyang Zhao

Public academic CV | November 2026

zhaojingyang@westlake.edu.cn

RESEARCH

Microfluidic systems and automation; biosensing and biological platforms; soft biointerfaces and robotics.

EDUCATION AND EXPERIENCE

2022-2026: Doctoral research, Materials Science and Engineering, Westlake University / Zhejiang University. Advisor: Hanqing Jiang.

2021-2022: M.S., Mechanical Engineering, Columbia University. BioMEMS Lab, Qiao Lin.

2019: Summer research visitor, University of Michigan. Katsuo Kurabayashi.

2016-2020: Zhejiang University, Mechanical Design, Manufacturing and Automation; second major in English.

RESEARCH CONTRIBUTIONS

Programmable microfluidic synthesis: Platform development and control of microfluidic reaction environments.

Integrated point-of-care diagnostics: Experimental investigation, analysis and device-workflow work; the qPCR and pneumatic studies retain their separate evidence.

Quantitative biosensing: Contributions to microfluidic assays, device integration and image analysis across collaborative studies.

Soft biointerfaces & wearable sensing: Origami prototype fabrication for haptics, tissue–probe mechanical analysis, and epidermal sensor development.

Soft robotics & microfabrication: MEMS fabrication for insect-scale robotics and SEM characterization for biodegradable origami systems.

PUBLICATIONS

Discrimination for fifteen genotypes of human papillomavirus (HPV) via multiplexed qPCR in a portable and automated device. Sensors and Actuators B: Chemical, 2026. https://doi.org/10.1016/j.snb.2026.140134

Continuous Synthesis of Optimized Ni3(HITP)2 With Broad Spectrum of Morphologies via Automated Microfluidic Ammonia Manipulation Platform. Small Methods, 2026. https://doi.org/10.1002/smtd.70746

An energy-embodied paralleled liquid manipulation for equipment-free, quantitative multiplexed liver function monitoring. Science Advances, 2025. https://doi.org/10.1126/sciadv.adx0092

Muscle-inspired elasto-electromagnetic mechanism in autonomous insect robots. Nature Communications, 2025. https://doi.org/10.1038/s41467-025-62182-2

Biodegradable origami enables closed-loop sustainable robotic systems. Science Advances, 2025. https://doi.org/10.1126/sciadv.ads0217

Bioelastic state recovery for haptic sensory substitution. Nature, 2024. https://doi.org/10.1038/s41586-024-08155-9

Implantation-assistance-free flexible waveguide probe for optogenetic stimulation. Cell Reports Physical Science, 2024. https://doi.org/10.1016/j.xcrp.2024.102217

Aptameric Metal–Organic Framework Nanobiosensor. ACS Applied Nano Materials, 2024. https://doi.org/10.1021/acsanm.4c03305

Assessing aptamer-analyte binding kinetics by microfluidic fluorescence microscopy. Sensors and Actuators B: Chemical, 2024. https://doi.org/10.1016/j.snb.2023.135018

Microfluidic measurement of intracellular mRNA with a molecular beacon probe towards point-of-care radiation triage. Sensors & Diagnostics, 2024. https://doi.org/10.1039/D4SD00079J

Materials and Device Design for Epidermal UV Sensors with Real■Time, Skin■Color Specific, and Naked■Eye Quasi■Quantitative Monitoring Capabilities. Advanced Materials Technologies, 2023. https://doi.org/10.1002/admt.202201481

Machine learning-based cytokine microarray digital immunoassay analysis. Biosensors and Bioelectronics, 2021. https://doi.org/10.1016/j.bios.2021.113088

METHODS

Soft lithography; photolithography; microfluidic device design; COMSOL; fluid-structure interaction; Python; MATLAB; image analysis.

Public summary based on the supplied CV and linked publication records. Degree conferral and availability are not asserted.